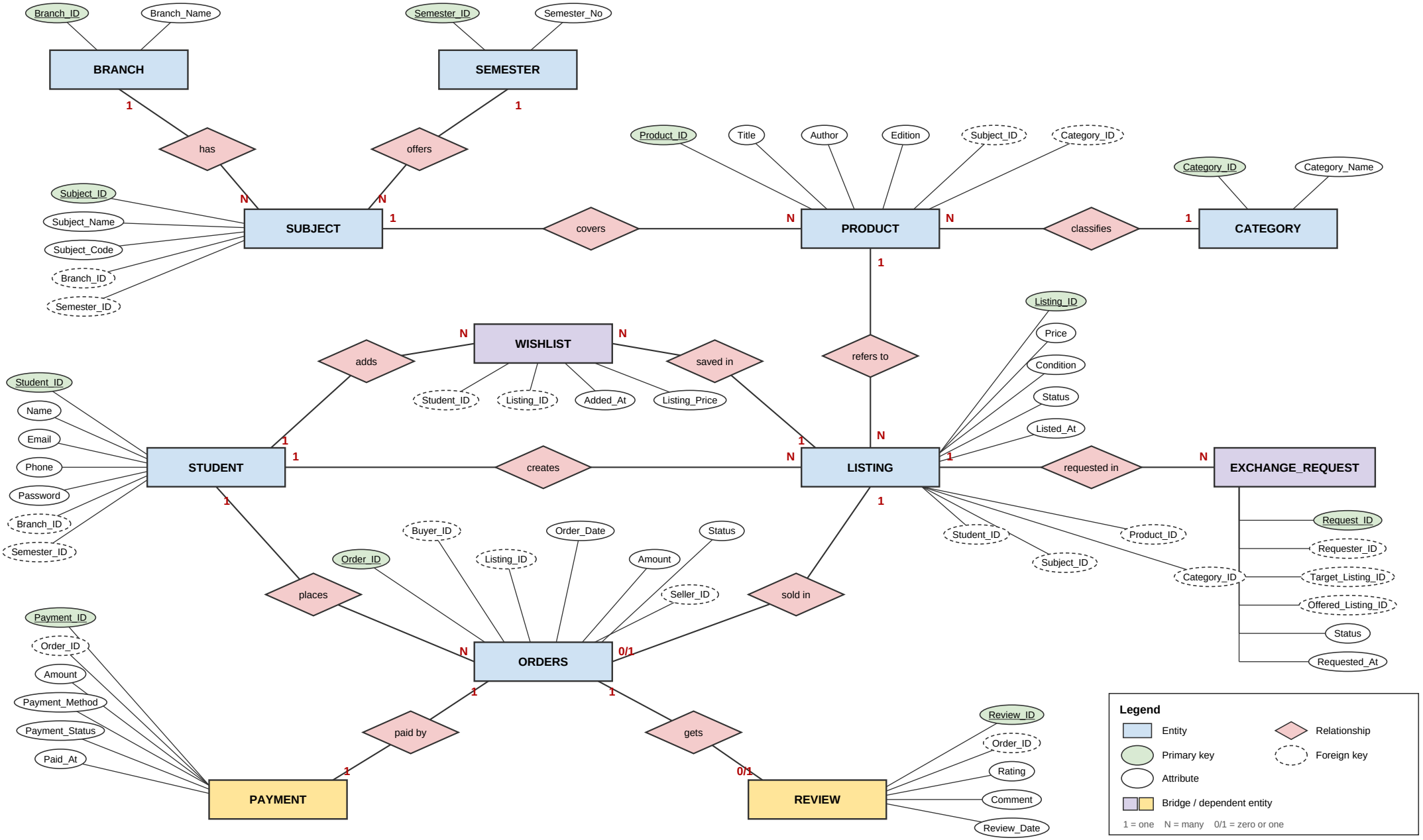
CAMPUSSWAP - STUDENT MARKETPLACE MANAGEMENT SYSTEM

Second-Hand Books and Study Equipment - ER Diagram, Relational Schema and Normalization

Name: Sharvari H D **USN:** U25UV24T029099 **Class:** 3rd Year, CSE **Subject:** DBMS

1. Entity-Relationship Diagram (before normalization)

STUDENT also references BRANCH and SEMESTER (Branch_ID, Semester_ID). EXCHANGE_REQUEST also stores the requester (STUDENT) and the offered LISTING.



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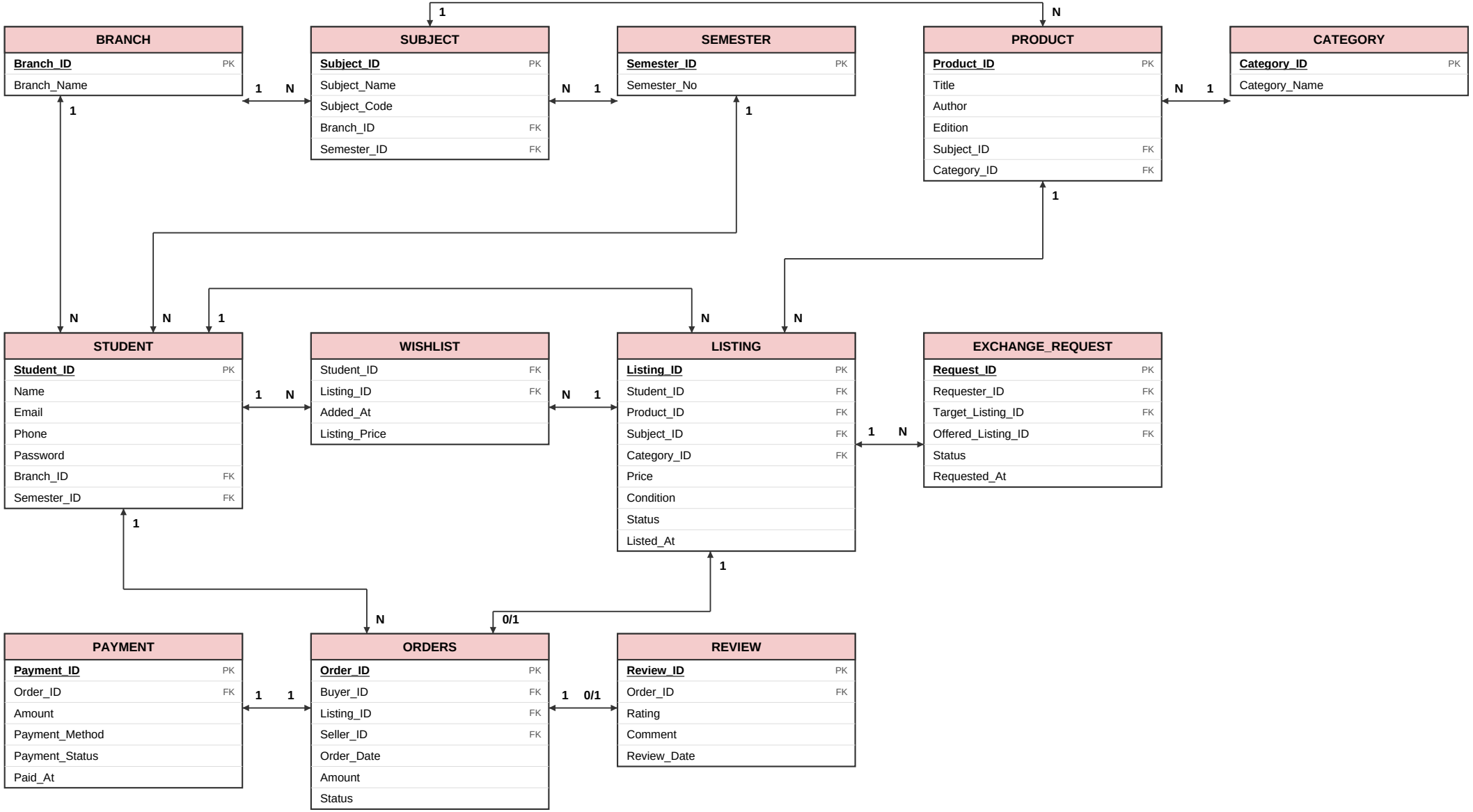
2. Relational Schema (before normalization)

The ER diagram is mapped to twelve relations. These are the tables checked in the normalization process that follows.



PK = primary key (underlined) FK = foreign key Arrows join the parent table (1 side) to the child table (N side). EXCHANGE_REQUEST.Requester_ID references STUDENT and Offered_Listing_ID references LISTING; ORDERS.Buyer_ID references STUDENT.

3. Normalization Process

Each of the twelve relations from Section 2 is tested against the normal forms in order, from 1NF to 5NF. Only three tables needed a change (WISHLIST, LISTING and ORDERS); the rest were already in 3NF.

1NF - First Normal Form	
Applied	• All attributes hold atomic (single) values; there are no repeating groups or multivalued attributes. • The many listings a student saves are stored as separate rows in the bridge table WISHLIST, and every copy a student sells is a separate row in LISTING, instead of storing lists in a single field of STUDENT. • **Change:** WISHLIST had no primary key (Student_ID and Listing_ID were only foreign keys), so the same listing could be saved twice by one student. The composite primary key (Student_ID, Listing_ID) is added.

2NF - Second Normal Form	
Applied	• Checked only for tables with a composite key: WISHLIST. All other tables have a single-column primary key, so they cannot have a partial dependency. • WISHLIST(Student_ID, Listing_ID) -> Added_At depends on the whole key (when this student saved this listing). • **Change:** Listing_Price depended only on Listing_ID (a partial dependency) and was repeated for every student who saved the same listing, causing update anomalies. It is removed; the price is read from LISTING.Price.

3NF - Third Normal Form	
Applied	• **Change:** LISTING stored Subject_ID and Category_ID. Since Listing_ID -> Product_ID -> Subject_ID, Category_ID, this is a transitive dependency. Both are removed and are obtained through PRODUCT. • **Change:** ORDERS stored Seller_ID. Since Order_ID -> Listing_ID -> Student_ID (the seller), this is transitive. Seller_ID is removed; the seller is found through LISTING.Student_ID. • Subject_Name and Subject_Code depend only on Subject_ID; Branch_Name only on Branch_ID; Title, Author and Edition only on Product_ID; Name, Email, Phone and Password only on Student_ID. • Amount in ORDERS is kept as a stored value (the price at the time of sale), so it depends only on Order_ID.

4NF - Fourth Normal Form	
N/A	• No independent multivalued dependencies remain: a student's wishlist and a student's own listings are already stored in the separate tables WISHLIST and LISTING. • Hence 4NF needs no further change.

5NF - Fifth Normal Form	
N/A	• No non-trivial join dependencies remain. The many-to-many STUDENT-LISTING relationship is resolved by WISHLIST, and EXCHANGE_REQUEST stores the target and offered listings in one row. • Hence 5NF needs no further change.

Summary of changes

Table	Before	After	Normal form
WISHLIST	Student_ID and Listing_ID were foreign keys only; no primary key	Composite PK (Student_ID, Listing_ID), both also FK	1NF
WISHLIST	Listing_Price stored (depends only on Listing_ID)	Listing_Price removed; read from LISTING	2NF
LISTING	Subject_ID and Category_ID stored (via Product_ID)	Both removed; read through PRODUCT	3NF
ORDERS	Seller_ID stored (via Listing_ID)	Seller_ID removed; seller = LISTING.Student_ID	3NF
All other tables	-	Unchanged; already in 3NF	3NF

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4. Final Normalized ER Diagram (3NF)

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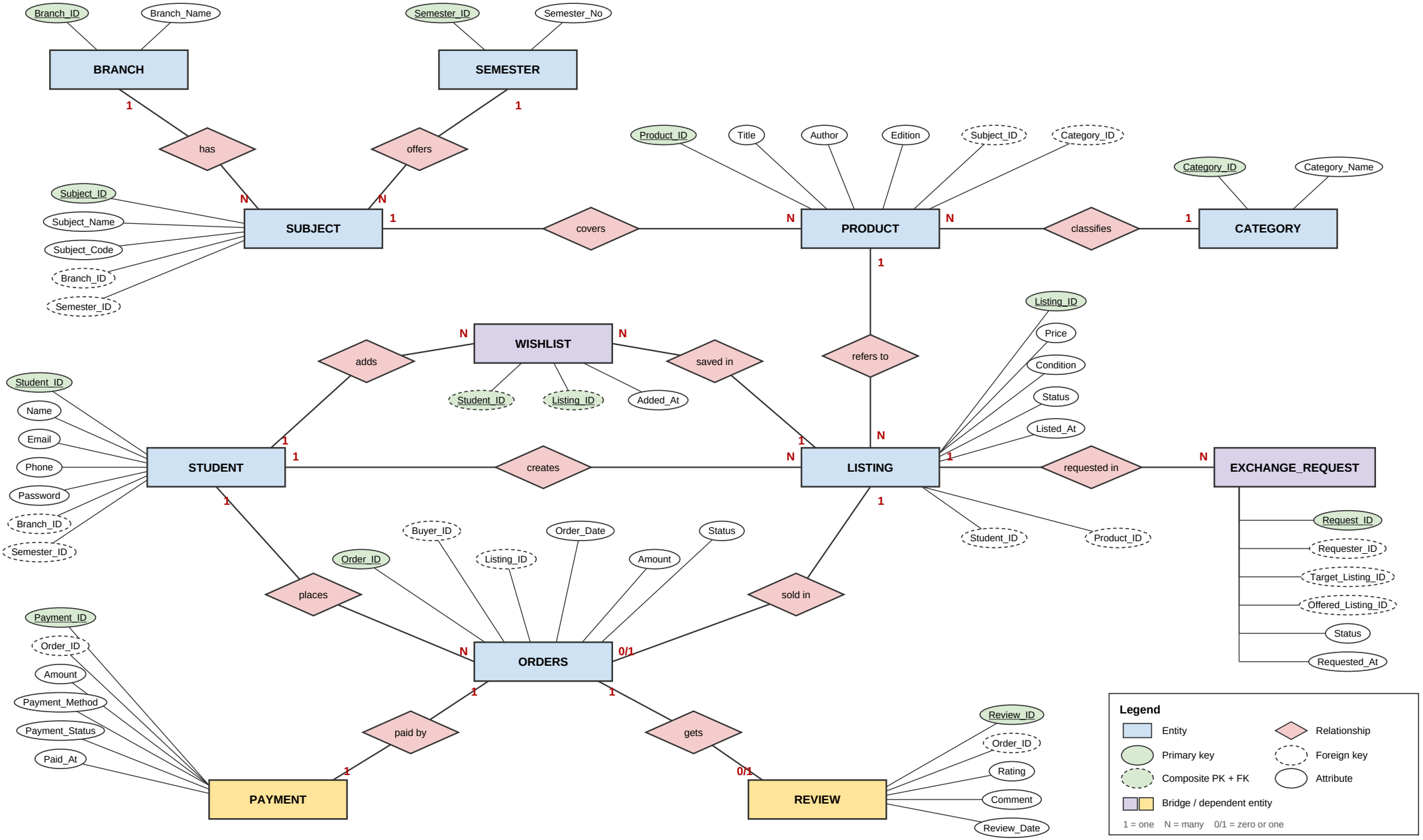

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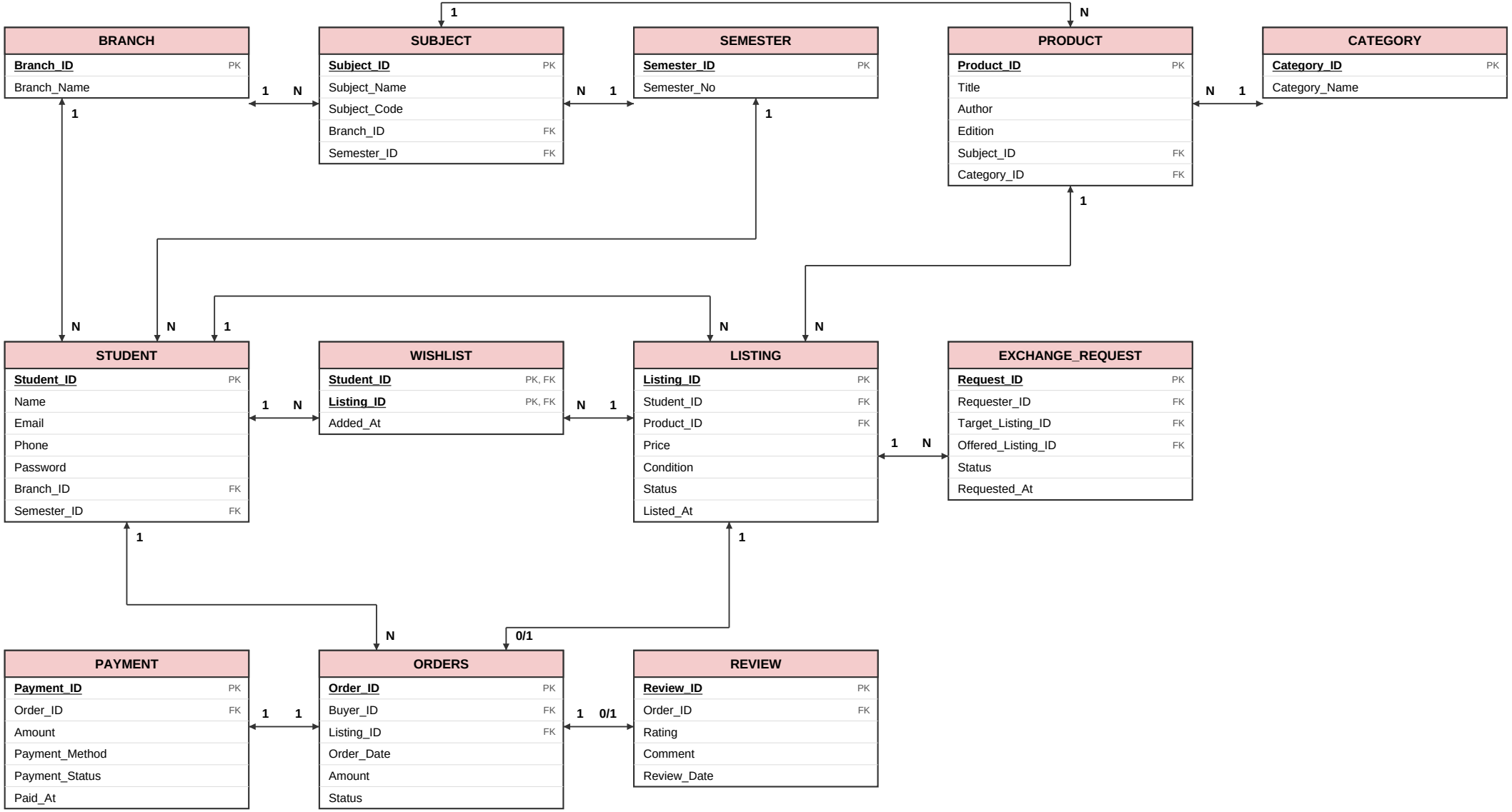
5. Final Normalized Relational Schema (3NF)

WISHLIST now uses a composite primary key and no longer stores Listing_Price; Subject_ID and Category_ID are removed from LISTING and Seller_ID from ORDERS. All other tables are unchanged.



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